

# FINEXA AlphaPack™ (Grey Wolf Optimizer) — User-Friendly Guide & Technical Specification

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## 1. What is this Module? (In Simple Words)

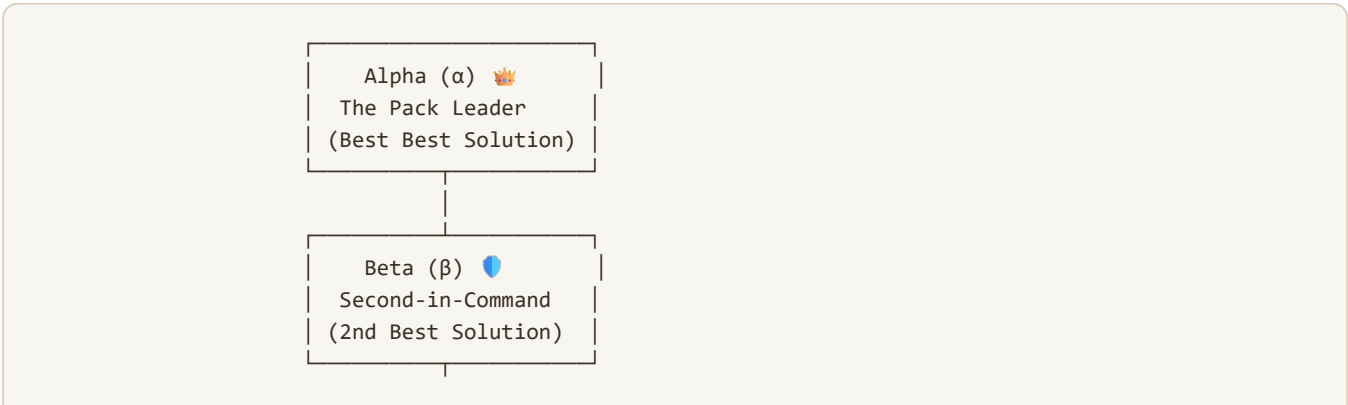
**FINEXA AlphaPack™** is an AI-powered smart investment optimizer that finds the absolute best place to invest your monthly savings.

When you want to invest (for a car, home down payment, child education, or retirement), you have many choices: \* Large Cap Stocks (Nifty 50) \* Mid & Small Cap High-Growth Stocks \* Safe Corporate Bonds & Debt \* Sovereign Gold / Gold ETFs \* Global / US Tech Index

Instead of guessing how much percentage to put in each, **FINEXA releases a virtual pack of 25 AI wolves.** These wolves hunt through thousands of market combinations to find the exact combination that gives you the **highest return with the lowest risk.**

## 2. The Nature Concept: How Grey Wolves Hunt

In nature, grey wolves live in organized packs with strict social hierarchy:





- 1. **Alpha ( $\alpha$ ):** The dominant leader. In FINEXA, this is the **Champion Portfolio** that achieves the highest financial score.
- 2. **Beta ( $\beta$ ):** The trusted advisor. In FINEXA, this is the **Defensive Shield Portfolio** (lower risk, steady safety).
- 3. **Delta ( $\delta$ ):** The scouts. In FINEXA, this is the **Aggressive Growth Portfolio** (captures maximum upside).
- 4. **Omega ( $\omega$ ):** All other wolves follow the direction of  $\alpha, \beta, \delta$  towards the prey (the global optimal investment balance).

### 3. The 3 Pack Leaders: Alpha, Beta, Delta

When you run FINEXA AlphaPack™, it instantly presents the top 3 leader strategies:

| LEADER             | TITLE          | ROLE & STYLE                                                                              | WHO IT'S BEST FOR                                                 |
|--------------------|----------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Alpha ( $\alpha$ ) | 👑 The Champion | <b>Global Best Balance:</b> Maximizes the Sharpe ratio (highest return per unit of risk). | Everyday investors seeking balanced, steady wealth compounding.   |
| Beta ( $\beta$ )   | 🛡️ The Shield  | <b>Capital Preservation:</b> Heavy bonds and gold buffer with lower volatility.           | Short-term goals (1–3 yrs) or low risk comfort.                   |
| Delta ( $\delta$ ) | 🏹 The Hunter   | <b>Aggressive Alpha:</b> Heavy high-growth mid-caps and global tech equities.             | Long-term goals (10–15+ yrs) targeting maximum financial freedom. |

### 4. How the GWO Algorithm Works in Finance

#### 4.1 Mathematical Equations of the Pack (Mirjalili et al., 2014)

During the hunt, wolves encircle the prey according to two vectors:

$$\vec{D} = |\vec{C} \cdot \vec{X}_p(t) - \vec{X}(t)|$$
$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \cdot \vec{D}$$

Where: \*  $\vec{X}(t)$  is the current position vector (portfolio weights across the 5 asset classes). \*  $\vec{X}_p(t)$  is the position of the prey (the optimal portfolio). \*  $\vec{A} = 2\vec{a} \cdot \vec{r}_1 - \vec{a}$  and  $\vec{C} = 2\vec{r}_2$  are coefficient vectors. \* Parameter  $\vec{a}$  linearly decreases from 2.0 to 0 over iterations to balance **Exploration (searching for prey)** and **Exploitation (attacking prey)**.

4.2 Updating Position via the 3 Leaders (\$\alpha\$, \$\beta\$, \$\delta\$)

For every Omega wolf \$i\$, its new position is guided by the 3 leaders:

$$\vec{D}\alpha = |\vec{C}_1 \cdot \vec{X}\alpha - \vec{X}i|, \quad \vec{X}_1 = \vec{X}\alpha - \vec{A}_1 \cdot \vec{D}\alpha$$
$$\vec{D}\beta = |\vec{C}_2 \cdot \vec{X}\beta - \vec{X}i|, \quad \vec{X}_2 = \vec{X}\beta - \vec{A}_2 \cdot \vec{D}\beta$$
$$\vec{D}\delta = |\vec{C}_3 \cdot \vec{X}\delta - \vec{X}i|, \quad \vec{X}_3 = \vec{X}\delta - \vec{A}_3 \cdot \vec{D}\delta$$
$$\vec{X}_{i(t+1)} = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3}$$

4.3 The Fitness Function (Sharpe Ratio & Return)

The fitness score of each portfolio position \$\vec{w} = [w\_1, \dots, w\_5]\$ is evaluated by:

$$\text{Expected Return } E[R_p] = \sum_{i=1}^5 w_i \cdot \mu_i$$
$$\text{Portfolio Variance } \sigma_p^2 = \vec{w}^T \cdot \mathbf{\Sigma} \cdot \vec{w}$$
$$\text{Sharpe Ratio } S_p = \frac{E[R_p] - R_f}{\sigma_p}$$
$$\text{Fitness}(\vec{w}) = 4.0 \cdot S_p + 2.5 \cdot E[R_p] - 2.0 \cdot \sigma_p$$

5. Interactive Features in the App

- 1. **Step 1: Monthly Investment Slider:** Choose your SIP amount (₹2,000 to ₹1,00,000/month).
- 2. **Step 2: Goal Time Horizon:** Select 1 to 20 years with quick shortcuts (3y Car, 5y Home, 10y Education, 15y Retirement).
- 3. **Step 3: Risk Comfort:** Choose Safe & Steady 🟢, Balanced 🟡, or Aggressive 🔴.
- 4. **Step 4: Pick Your Leader:** Switch between \$\alpha\$, \$\beta\$, and \$\delta\$ with 1 tap.
- 5. **Exact Rupee Breakdown:** Shows exactly how many rupees to put in Nifty 50, Mid Caps, Debt, Gold, and Global Tech every month.
- 6. **Future Wealth Growth Card:** Projects your exact terminal corpus value and total profit.

6. Real-World Example (₹15,000/month for 5 Years)

User Input:

- **Monthly SIP:** ₹15,000 / month
- **Horizon:** 5 Years (60 Months)
- **Risk:** Balanced Growth

👑 Alpha Leader Output:

- **Expected Return:** 13.8% / year
- **Portfolio Volatility (Risk):** 12.4% (Moderate)
- **Sharpe Ratio:** 1.15

Monthly Asset Split:

- **Large Cap (Nifty 50):** 40% = ₹6,000/mo
- **Mid & Small Cap Growth:** 20% = ₹3,000/mo

-  **Corporate Bonds & Debt:** 20% = ₹3,000/mo
-  **Sovereign Gold ETF:** 10% = ₹1,500/mo
-  **Global / US Tech Index:** 10% = ₹1,500/mo

 **5-Year Wealth Result:**

- **Total Money Invested:** ₹9,00,000
- **Estimated Future Wealth:** ₹12,85,400
- **Total Profit Gained:** +₹3,85,400 (+1.43x Multiple) 

## 7. Why GWO Beats Traditional Portfolio Calculators

| DIMENSION                | TRADITIONAL FIXED RULES                                                                               | BASIC SIP CALCULATOR                                                                                      | FINEXA ALPHAPACK™ (GWO)                                                                                                                                                                  |
|--------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Multi-Asset Intelligence |  None                |  Single asset assumption |  <b>5-Asset Covariance Balancing</b>                                                                  |
| Optimization Method      |  Guesswork           |  Static compound formula |  <b>Grey Wolf Pack Swarm Optimization</b>                                                             |
| Leader Choices           |  One-size-fits-all |  Single number         |  <b>3 Leaders (<math>\alpha</math>, <math>\beta</math>, <math>\delta</math>) to fit personality</b> |
| Downside Protection      |  Ignored           |  Ignored               |  <b>Bonds &amp; Gold Volatility Dampener</b>                                                        |
| User Friendliness        |  Confusing         |  Basic                 |  <b>Clean cards, sliders &amp; plain-English advice</b>                                             |

Report Generated for FINEXA AI — Autonomous Wealth Systems.